



Influence of strain rate and temperature on tensile properties and flow behaviour of a reduced activation ferritic–martensitic steel

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ABSTRACT

Tensile strength and flow behaviour of a Reduced Activation Ferritic–Martensitic (RAFM) steel (9Cr–1W–0.06Ta–0.22V–0.08C) have been investigated over a temperature range of 300–873 K at different strain rates. Tensile strength of the steel decreased with temperature and increased with strain rate except at intermediate temperatures. Negative strain rate sensitivity of flow stress of the steel at intermediate temperatures revealed the occurrence of dynamic strain ageing in the steel, even though no serrated flow was observed. The tensile flow behaviour of the material was well represented by the Voce strain hardening equation for all the test conditions. Temperature and strain rate dependence of the various parameters of Voce equation were interpreted with the possible deformation mechanisms. The equivalence between the saturation stress at a given strain rate in tensile test and steady state deformation rate at a given stress in creep test was found to be satisfied by the RAFM steel.

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1. Introduction

In nuclear fusion reactor applications, the blanket module facing the plasma is expected to be exposed to the simultaneous effects of mechanical and thermo-mechanical stresses and intense neutron irradiation-induced damage at high temperatures. Ferritic steels as structural materials for the blanket module of fusion reactors are now under prime consideration because of their inherent void swelling resistance coupled with low thermal expansion coefficient and high thermal conductivity compared to austenitic steels [1,2]. The requirement of safe operation and decommissioning of blanket module of fusion power plants and the disposal of radioactive wastes have also demanded the development of steels with reduced residual radioactive characteristics [3,4]. High chromium ferritic–martensitic steels have been used extensively in power generation and petrochemical industries. The RAFM steels are modelled after the well established modified 9Cr–1Mo steel by replacing the highly radioactive elements residual molybdenum and niobium respectively with tungsten and tantalum having comparatively lower residual radioactivity. The steel is generally used in the normalised and tempered condition and has a tempered martensitic structure. It derives its high temperature strength from the complex microstructures consisting of high dislocation density, sub-boundaries decorated with $M_{23}C_6$ carbides and (Ta,V)-carbides of MX type in the intragranular region [5–7].

To predict the performance of the reactor components, an understanding of the plastic flow behaviour of the structural materials is essential. In the present work, detailed analysis of tensile deformation of a RAFM steel has been carried out to understand the high temperature plastic flow behaviour of the material. The plastic flow behaviour of the material has been studied in terms of the constitutive relationships proposed by Hollomon [8], Ludwik [9], Ludwigson [10] and Voce [11]. The appropriate constitutive relationship describing the tensile flow behaviour of the material over wide ranges of strain rate and temperature has been identified. The parameters of the relevant constitutive relationship have been examined in detail in order to gain an insight into the variations in strain-hardening characteristics of the steel at different temperatures and strain rates.

2. Experimental

The RAFM steel has been produced at M/s. Mishra Dhatu Nigam Limited (MIDHANI), Hyderabad, India. The ingot was produced by vacuum induction melting followed by vacuum arc refining processes. The ingot was hot-forged and subsequently rolled to a 12 mm thick plate. The steel plates were subjected to the final heat treatment consisting of normalisation at 1250 K for 30 min followed by tempering at 1033 K for 90 min. The chemical composition of the steel is shown in Table 1.

Button-head cylindrical specimens having gauge length of 20 mm and gauge diameter of 4 mm were machined from the

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Table 1

Chemical composition (wt.%) of the RAFM steel.

Element	Cr	C	Mn	V	W	Ta	N	O	P	S
wt.%	9.04	0.08	0.55	0.22	1.01	0.06	0.0226	0.0057	0.002	0.002
Element	B	Ti	Nb	Mo	Ni	Cu	Al	Si	Co	As + Sn + Sb + Zr
wt.%	0.0005	<0.005	0.001	0.001	0.005	0.001	0.004	0.09	0.004	<0.03

heat-treated plates. The gauge length of the specimen was parallel to the rolling direction of the plate. Tensile tests were carried out employing a screw-driven tensile testing machine fitted with furnace and digital data acquisition system. Tensile tests were conducted in air at nominal strain rates of $3 \times 10^{-3} \text{ s}^{-1}$, $3 \times 10^{-4} \text{ s}^{-1}$ and $3 \times 10^{-5} \text{ s}^{-1}$ at temperatures ranging from 300 to 873 K at intervals of 50 K. The temperature during testing was controlled within ± 1 K.

3. Results and discussion

3.1. Tensile properties

Engineering stress–strain curves of the material tested at a strain rate of $3 \times 10^{-3} \text{ s}^{-1}$ over the temperature range of 300–873 K are shown in Fig. 1. Similar variation in stress–strain was also observed at other two investigated strain rates. The results revealed that the material exhibited monotonic stress–strain curves over the entire temperature range with no serrated flow in the intermediate temperature range as commonly observed in ferritic steels [12]. The strain to failure (ϵ_f) was found to reduce gradually with increase in temperature from 300 to around 723 K, beyond which it increased.

The variations in 0.2% offset yield stress (YS) and ultimate tensile strength (UTS) of the steel with test temperature at various strain rates are shown in Figs. 2 and 3 respectively. Tensile strength of the steel decreased with increase in test temperature. The rate of decrease of strength with temperature was relatively lower at the intermediate temperature range of 423–673 K, beyond which the rate of decrease was higher. Tensile strength of the steel was found to increase with strain rate in general. The percent elongation and percent reduction of area (Figs. 4 and 5) showed lower values in the intermediate temperature range, followed by a relatively rapid increase above 623 K. A plateau or slower rate of the variation in tensile strength with temperature or even peak in tensile strength in the intermediate temperature range in steels has been reported to be associated with the occurrence of dynamic strain ageing (DSA) [13]. Dynamic strain ageing occurs within a certain range of temperatures and strain rates which facilitates dynamic interaction between mobile dislocations and solute atoms. Several manifestations of DSA were observed with different materials [13]. The most commonly observed form of DSA is serrations in the tensile flow curve which appears due to repeated locking and unlocking of dislocations in solute atmospheres [13,14]. Serrated tensile flow was not observed in the presently investigated RAFM steel. Negative strain rate sensitivity of flow stress i.e. true stress at a particular true strain of the material is one of the manifestations of the occurrence of DSA [13]. Strain rate sensitivity test on the steel at 300 K and 573 K was carried out by changing the strain rate during the tensile test. The strain rate change response of the steel to the flow stress at 300 K and 573 K is shown in Fig. 6. Negative strain rate sensitivity to flow stress was observed during tensile test at 573 K against positive strain rate sensitivity to flow stress at 300 K. This confirmed decisively that the dynamic strain ageing occurred in this steel also. Presence of weak serrated flow in the stress–strain curves has been reported by Li et al. [15] in JLF-I, a RAFM steel, in the intermediate temperature range. Slower rate of decrease in tensile strength (Figs. 2 and 3) of the steel in the

intermediate temperature range along with the lower ductility (Figs. 4 and 5) has been attributed to the occurrence of dynamic strain ageing in the steel.

3.2. Tensile flow behaviour

Strain hardening is the resistance to plastic flow of a material. This property depends on the inherent characteristics of the material and is reflected in the shape of the stress–strain curve from yield stress to ultimate tensile strength. The shape of the true stress vs. true strain curve from the onset of yielding to maximum force has been described empirically by several relations. In the present investigation, the true stress and true strain data of the steel obtained by performing tensile tests at different temperatures and strain rates have been analysed by computer in terms of the constitutive relations proposed by Holloman [8], Ludwik [9], Ludwigson [10] and Voce [11]. The true stress and true strain data were fitted to the constitutive equations using the Levenberg–Marquardt least square method with unknown constants as free parameters. The best fit is indicated by the lowest value of chi square (χ^2) which is the sum of the squares of the deviations of calculated stress values from the experimental data and by the smallest error band in determining the constants of the constitutive equations.

In the present investigation, the constitutive equations used to analyse the tensile flow behaviour of the steel are:

1. Hollomon [8]:

$$\sigma = K_H \epsilon^{n_H} \quad (1)$$

where σ is the true-stress, ϵ is the true-strain, K_H the strength coefficient and n_H the strain hardening exponent.

2. Ludwik [9]:

$$\sigma = \sigma_0 + K_L \epsilon^{n_L} \quad (2)$$

where σ_0 is a material constant which accounts for the positive deviation owing to yielding at low strains, K_L is the strength coefficient and n_L the strain hardening exponent.

3. Ludwigson [10]:

$$\sigma = K_1 \epsilon^{n_1} + \exp(K_2 + n_2 \epsilon) \quad (3)$$

where K_1 and n_1 are the strength coefficient and strain hardening exponent respectively, and K_2 and n_2 are additional constants.

4. Voce [11]: Voce's equation can be used for analysing tensile flow behaviour when the stress tends to saturate at large strains.

$$\sigma = \sigma_s - (\sigma_s - \sigma_i) \exp \left[-\frac{(\epsilon - \epsilon_i)}{\epsilon_c} \right] \quad (4)$$

where σ_s is the saturation stress, σ_i is the stress at the start of plastic strain ϵ_i , the value of ϵ is usually set to 0 and ϵ_c is a characteristic strain constant. By setting $n_v = -1/\epsilon_c$, Eq. (4) reduces to

$$\sigma = \sigma_s - (\sigma_s - \sigma_i) \exp(n_v \epsilon) \quad (5)$$

All the above mentioned relationships correspond to a typical shape of true stress vs. true strain curve. The equation which follows the experimental curve with a good accuracy represents the material's tensile flow behaviour. The Hollomon, Ludwik and Ludwigson constitutive equations predict monotonic unbounded

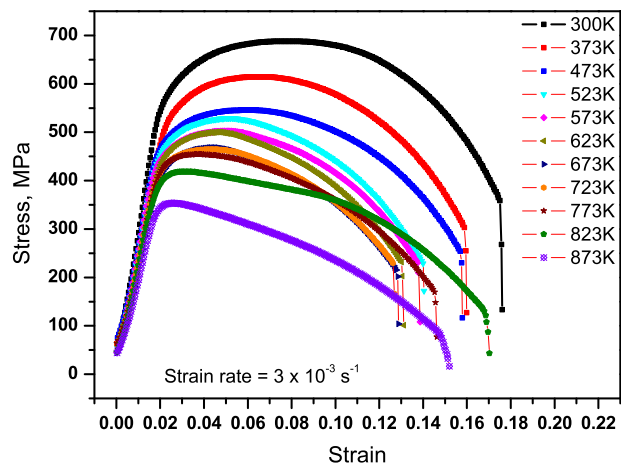


Fig. 1. Tensile curves of the RAFM steel at different test temperatures at a strain rate of $3 \times 10^{-3} \text{ s}^{-1}$.

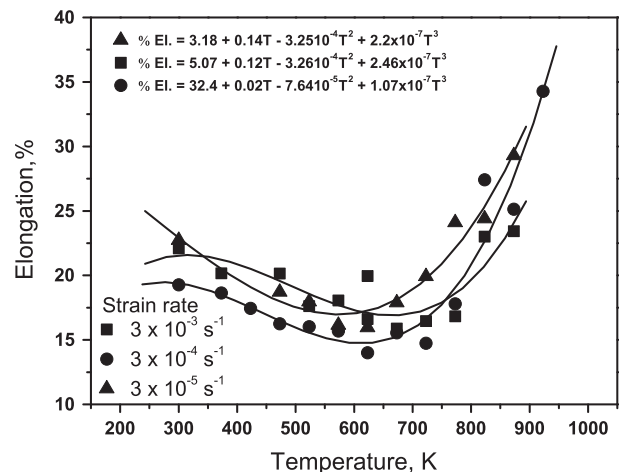


Fig. 4. Variation in percent elongation with test temperature at different strain rates.

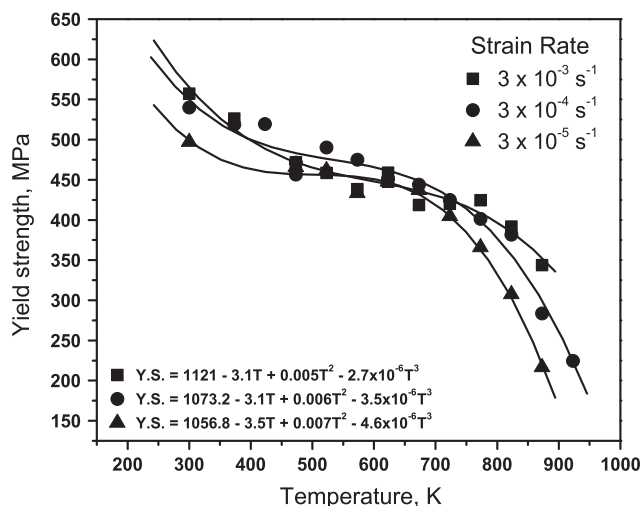


Fig. 2. Variation of yield stress of the steel with temperature at different strain rates.

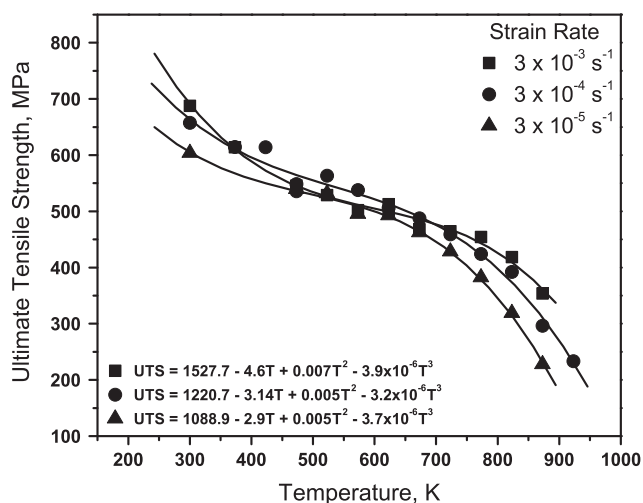


Fig. 3. Variation of ultimate tensile strength of the steel with temperature at different strain rates.

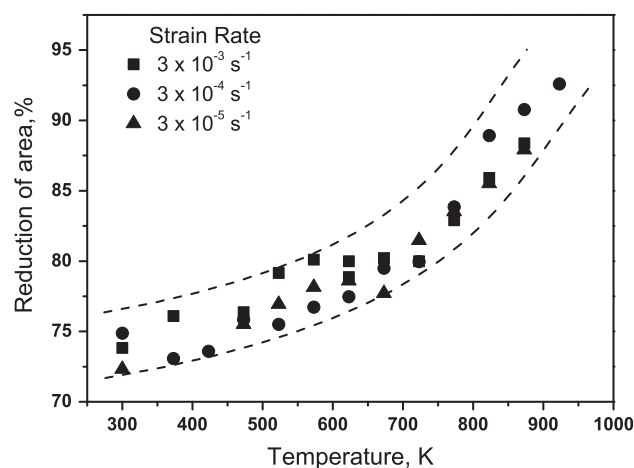


Fig. 5. Variation in percent reduction of area with temperature at different strain rates.

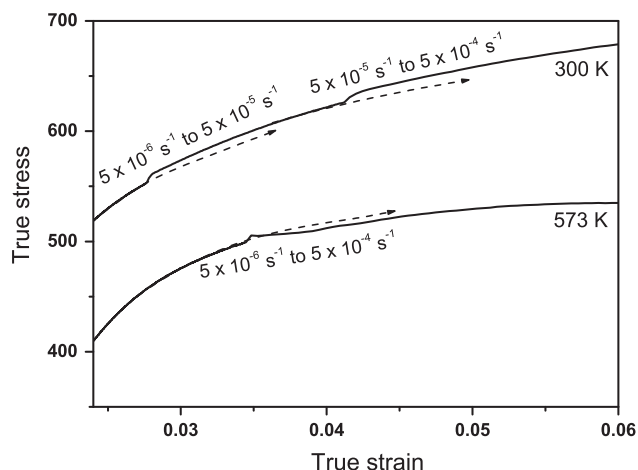


Fig. 6. Segments of true stress–true strain curve, showing strain rate sensitivity to flow stress of the steel at 300 K and 573 K. (The arrows show the likely progress of the flow behaviour of the material under the same strain rate condition. At 300 K, increase in flow stress was observed with the increase in strain rate; whereas at 573 K, decrease in flow stress was observed with the increase in strain rate.)

increase in flow stress as strain progresses, whereas the Voce constitutive equation presumes the existence of a saturation stress

at large strains. The true stress vs. true strain plots up to the ultimate tensile strength exhibited by the steel at 300 K and at

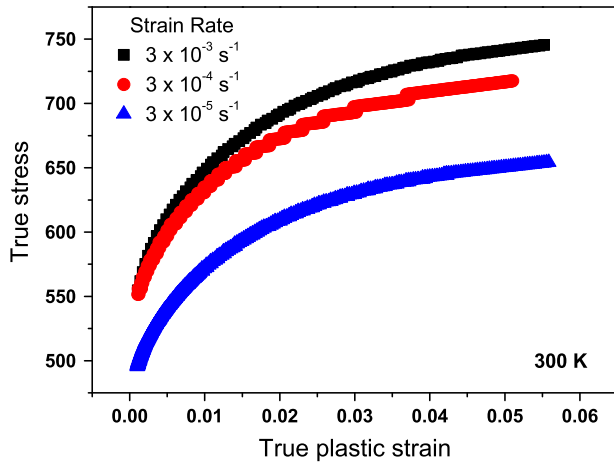


Fig. 7. Variation in true stress and true strain (plastic) of the steel at 300 K and at different strain rates.

different strain rates are shown in Fig. 7. Similar variations were obtained at other test temperatures and strain rates. The shape of the true stress and true strain curves with the tendency of saturation of stress at large strains indicated that the stress–strain data could be best fitted by the Voce constitutive equation than by the other constitutive equations. In this analysis, initially all the above mentioned relationships have been fitted to the experimental stress–strain data of the tensile tests at all test temperatures and strain rates. The values of χ^2 and minimum error band in determining the parameters of the different constitutive equations are compared in Table 2 for tensile tests carried out at 300 K and 823 K and at a strain rate of $3 \times 10^{-3} \text{ s}^{-1}$. Similar observations were also obtained for tensile tests at other temperatures and strain rates. At all the test temperatures and strain rates, the true stress and true strain data were adequately described by the Voce equation with the lowest values of χ^2 and minimum error band in determining the parameters in the constitutive equation (Table 2). The best description of tensile flow behaviour of the ferritic–martensitic RAFM steel by the Voce equation is in agreement with those reported in the literature for other ferritic steels [12,16,17].

3.2.1. Saturation stress σ_s in Voce constitutive equation

The temperature dependence of the saturation stress σ_s in Voce constitutive equation at different strain rates is shown in Fig. 8. The saturation stress σ_s at all the strain rates decreased with

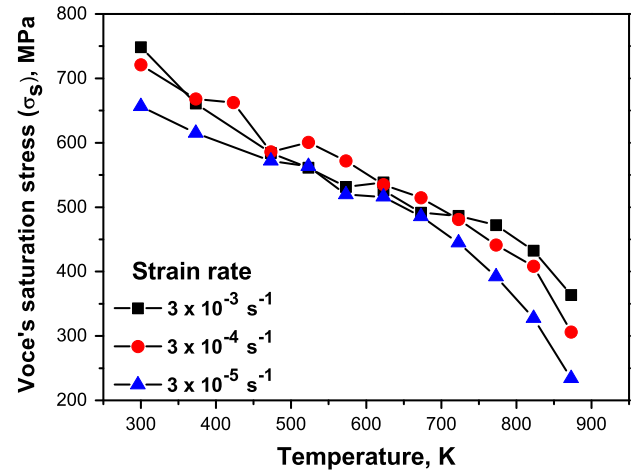


Fig. 8. Variation of Voce's parameter saturation stress σ_s with temperature at different strain rates.

temperature with relatively lower rates in the intermediate temperature range. The saturation stress was found to increase with strain rate. The variation of σ_s with temperature was similar to the variation of tensile strength with temperature (Fig. 3). The saturation stress σ_s represents a dynamic equilibrium between the strain hardening where the dislocations generated lead to the increase in dislocation density and the recovery where decrease in dislocation density occurs as a result of the rearrangement and annihilation of dislocations by slip, climb and sub boundary migration [18]. Decrease in saturation stress with temperature suggests that recovery would have begun early during tensile deformation at higher temperature. The occurrence of necking in the specimen during tensile test terminated the saturation state of tensile deformation.

Strain hardening behaviour of the RAFM steel is shown in Fig. 9, describing the variations of strain hardening rate with flow stress. The tempered ferritic–martensitic RAFM steel displayed predominantly stage III hardening, where the strain hardening decreased almost linearly with true stress. In the stage III of hardening process, the rearrangement of dislocations occurs with strain, leading to the formation of dislocation cell structures through cross slip at homologous temperatures below approximately 0.5 and to the well-defined dislocation subgrain structure through dislocation climb at higher temperatures [19]. As the deformation progresses, these dislocation cells or subgrains become equiaxed and approach a stable size which depends on temperature and strain rate,

Table 2

Estimated values of the parameters in the different constitutive equations with the values of χ^2 and error band ($3 \times 10^{-3} \text{ s}^{-1}$ strain rate).

	Hollomon			Ludwik			Ludwigson			Voce		
300 K												
χ^2	10.396			10.03524			3.0961			6.86697		
R^2	0.99589			0.99607			0.9988			0.99731		
300 K		Value	Error		Value	Error		Value	Error		Value	Error
	K_H	953.67	± 1.97	σ_0	152.94	± 55.13	K_1	1225.4	± 403.8	σ_s	748.56	± 0.78
	n_H	0.083	± 0.00	K_L	816.98	± 47.31	n_1	0.38	± 0.08	σ_1	555.53	± 1.08
				n_L	0.107	± 0.01	K_2	6.13	± 0.05	n_v	−62.31	± 0.92
							n_2	−5.53	± 3.03			
823 K												
χ^2	8.49622			8.81003			0.05557			0.31712		
R^2	0.94836			0.94851			0.99969			0.99815		
823 K		Value	Error		Value	Error		Value	Error		Value	Error
	K_H	519.31	± 5.028	σ_0	−11.05	± 1489	K_1	1403.9	± 30.74	σ_s	432.32	± 0.21
	n_H	0.04	± 0.002	K_L	530.05	± 1451	n_1	0.3889	± 0.004	σ_1	365.28	± 1.07
				n_L	0.0396	± 0.138	K_2	5.71	± 0.008	n_v	−381.9	± 8.23
							n_2	−42.67	± 1.766			

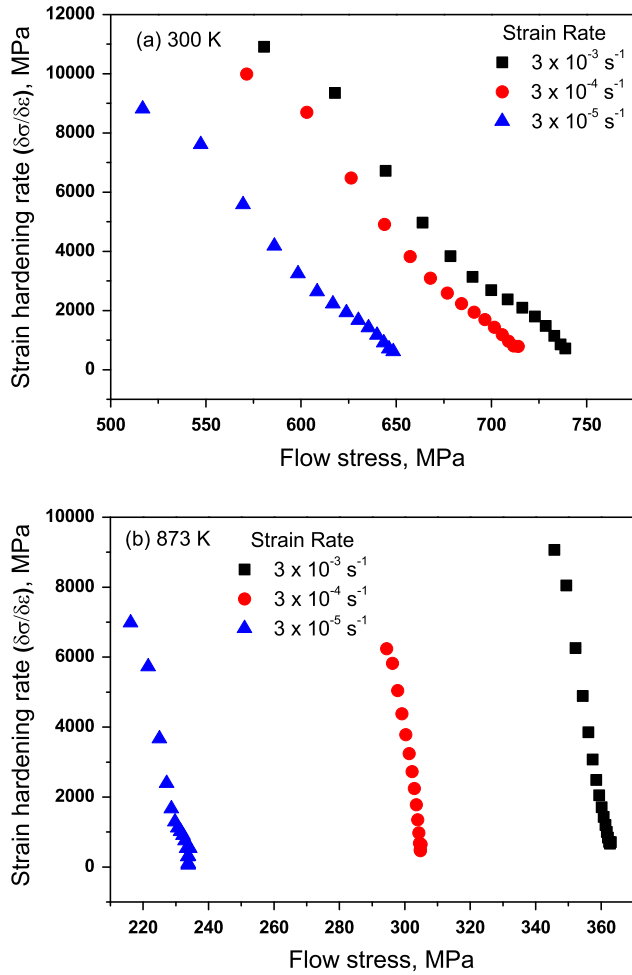


Fig. 9. Variations of strain hardening with flow stress of the material at different strain rates: (a) 300 K and (b) 873 K.

leading to a steady state condition. The saturation stress or the steady state stress σ_s has been related to the strain rate considering the thermally activated cross slip of dislocation as [20]

$$\dot{\epsilon}/\dot{\epsilon}_0 = (\sigma_s/\sigma_{s0})^{A/KT} \quad (6)$$

where $\dot{\epsilon}_0$ and σ_{s0} are constants and A is a parameter related to the stacking fault energy of the material. The σ_{s0} is related to the threshold stress for cross slip of dislocation. The variation of $\dot{\epsilon}$ with σ_s in log–log scale is shown in Fig. 10 for different testing temperatures. The relationship (Eq. (6)) was found to apply in the material both at low and high temperatures except in the intermediate temperature range of 423–673 K, implying that the mechanism controlling tensile deformation leading to a saturated state of stress is the cross slip of dislocation. Estrin and Mecking [18] had proposed that the above relation is valid for all temperatures, even at homologous temperatures above 0.5, where climb of dislocations is generally regarded as being the rate controlling process. In the DSA regime, dislocation movement would have been restricted to the plane and Eq. (6) would not be valid in these circumstances.

Strain hardening leads to increase in stress in a constant strain rate test (tensile test) and decrease in strain rate in constant stress test (creep test). It was proposed by Estrin and Mecking [18] that irrespective of the nature of deformation route (testing), the deformation mechanism remains similar, leading to an identical saturated state of dislocation substructure. The steady state creep

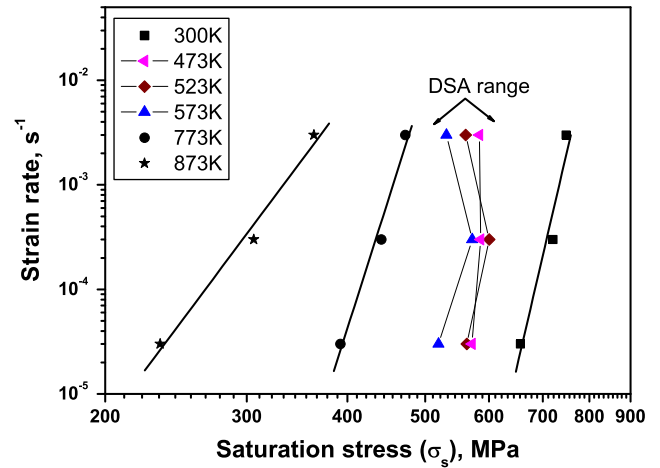


Fig. 10. Variation of saturation stress with strain rate at different temperatures.

rate ($\dot{\epsilon}_s$) with applied stress (σ_a) in the creep test can be described by a power law relation known as Norton's law:

$$\dot{\epsilon}_s = A_1 \sigma_a^n \quad (7)$$

where A_1 and n are constants.

Similarly, the relationship between saturated or steady state stress σ_s and applied strain rate ($\dot{\epsilon}_a$) in a tensile test can be described as

$$\dot{\epsilon}_a = A_2 \sigma_s^{\dot{n}} \quad (8)$$

where A_2 and $\dot{n}$ are constants.

Application of Eqs. (7) and (8) to the saturated/steady state tensile and creep deformation behaviour of the steel at 773 K, 823 K and 873 K is shown in Fig. 11. Creep deformation behaviour of the steel has been reported earlier [21,22]. It was found that $A_1 \approx A_2$ and $n \approx \dot{n}$. Excellent agreement between the steady creep rate at a given stress and the saturated stress at a given strain rate clearly showed that a steady microstructural state would have developed during deformation in microstructurally complex tempered martensitic RAFM steel [5–7,22].

3.2.2. Strain hardening parameter n_v in Voce's constitutive equation

The variation of Voce's constitutive parameter n_v with temperature at different strain rates is shown in Fig. 12. The variation

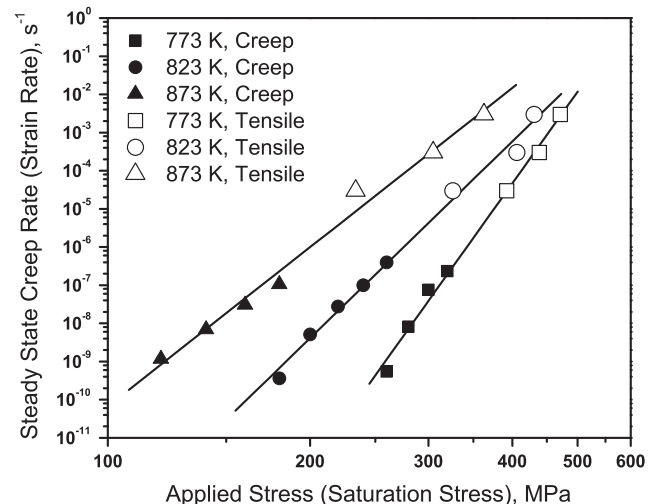


Fig. 11. Equivalence between saturation stress at a given strain rate in a tensile test and steady state creep rate at constant stress in a creep test.

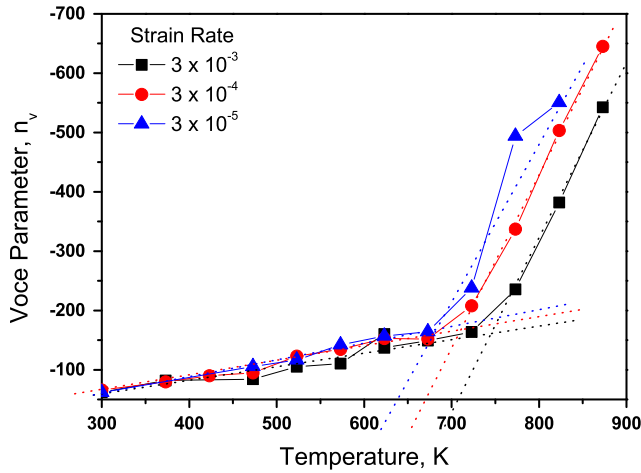


Fig. 12. Variation in Voce's parameter n_v with temperature at different strain rates.

shows a two-slope behaviour. The n_v value was found to increase marginally (in a negative scale) with temperature up to about 673 K beyond which it increased rapidly. Below around 673 K, the n_v values are not very sensitive to strain rate, whereas it is quite sensitive to strain rate beyond around 673 K. The temperature at which the change in slope of the temperature dependence of n_v occurred increased with strain rate. In literature, the values of n_v have been associated with various rate controlling mechanisms during tensile deformation. Gottstein and Argon [23] have shown that the decrease in strain hardening varies linearly with flow stress where recovery occurs as a result of cross-slip and other glide mechanisms, whereas in instances where climb and sub-boundary migration are the causes, the variation is found to be as the third power of flow stress. Thus, it is observed that n_v depends on the contributions of different recovery mechanisms. At low and medium temperatures where cross slip dominates, n_v is expected to have lower absolute values and at high temperatures where climb dominates, n_v should have very high absolute values. The transition from cross slip to climb, as revealed by the change in slope of n_v with test temperature (Fig. 12) was found to occur at lower temperature for slower strain rate tensile tests. This was probably because, during tensile deformation at lower strain rate, the dislocations had enough time to climb so that climb could occur at lower temperatures for lower strain rate tensile tests.

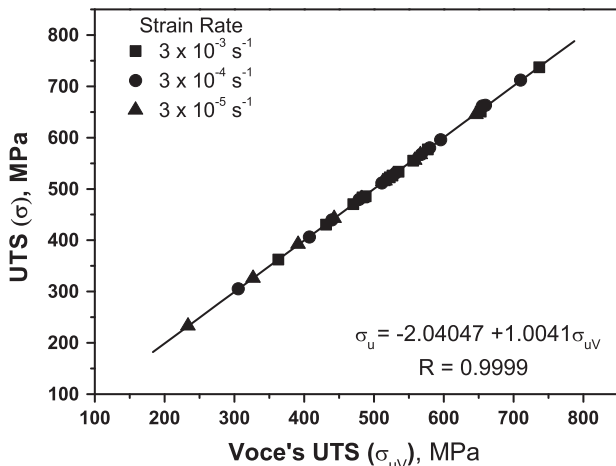


Fig. 13. Comparison between the experimentally determined true ultimate tensile stress and the value determined based on Voce's equation.

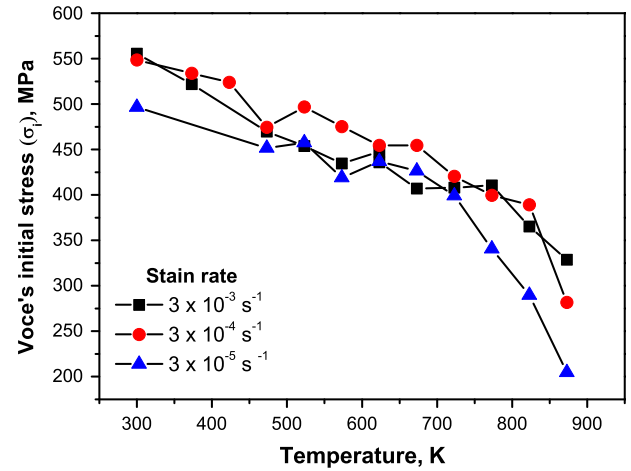


Fig. 14. Variation in Voce's parameter initial stress (σ_i) with temperature at different strain rates.

3.2.3. Relation between Voce's parameter and tensile strength

Ultimate tensile strength and saturated stress in Voce's constitutive equation can be correlated as both refer to the same phenomenon and they terminate at the onset of specimen necking during tensile deformation. Mishra et al. [24] have derived an expression, based on Voce's constitutive Eq. (5), to calculate ultimate tensile strength by invoking the Considère criterion of instability at the onset of necking.

On differentiating Eq. (5) and substituting $\exp(n_v \epsilon)$ with $(\sigma_s - \sigma)/(\sigma_s - \sigma_i)$ we get

$$\frac{\partial \sigma}{\partial \epsilon} = -n_v (\sigma_s - \sigma) \quad (9)$$

Now invoking Considère criterion ($\frac{\partial \sigma}{\partial \epsilon} = \sigma$) by substituting $\frac{\partial \sigma}{\partial \epsilon}$ and σ with σ_{uv} , we get

$$\sigma_{uv} = -n_v \sigma_s / (1 - n_v) \quad (10)$$

Fig. 13 compares the experimentally obtained true ultimate strength, σ_u (true stress corresponding to maximum engineering stress) and that obtained from Voce's constitutive equation parameters (σ_{uv}). It is observed that σ_u is linearly related to σ_{uv} with a correlation coefficient of 0.9999 as $\sigma_u = -2.04047 + 1.0041 \sigma_{uv}$.

The variation of Voce's constitutive equation parameter σ_i with temperature at different strain rates is shown in Fig. 14. The variation in initial stress σ_i with temperature is similar to that of saturation stress. The initial stress σ_i increased with increasing strain rate at low temperatures and high temperatures. However, at the intermediate temperatures, anomalous behaviour was observed. In Voce's constitutive equation (Eq. (4)), the initial plastic strain ϵ_i is set to zero to derive the modified version of Voce's constitutive equation (Eq. (5)). Initial stress is the stress at the start of the plastic strain and so it can be correlated with the yield stress of the material. Fig. 15 compares the initial stress σ_i with the yield stress of the material for all the tests carried out at different strain rates and temperatures. The straight line corresponds to $\sigma_i = YS$. The data follows the straight line. Thus it can be concluded that Voce's equation can determine the stress at the onset of plastic deformation.

The correspondence between the ultimate tensile strength and yield stress of the material with Voce's constitutive equation parameters (σ_s and σ_i) revealed that Voce's constitutive equation is a valid description of the tensile flow behaviour of the material over the temperatures and strain rates investigated.

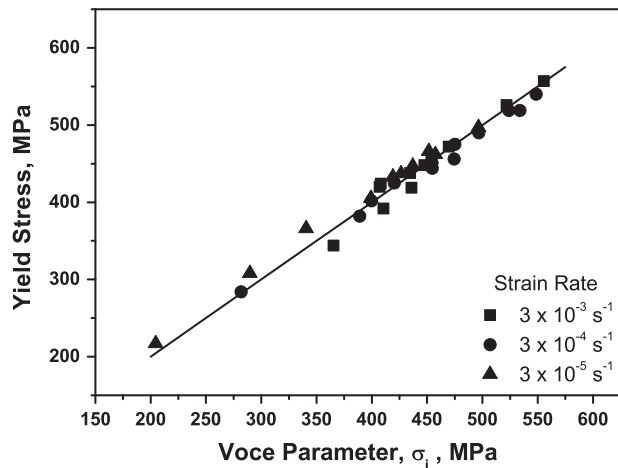


Fig. 15. Comparison between yield stress and Voce's parameter σ_i .

4. Conclusions

The following conclusions have been drawn from the investigation of tensile flow behaviour of the RAFM steel.

- Both the yield stress and ultimate tensile strength of the steel decrease with temperature with a relatively slower rate of decrease at the intermediate temperature range.
- Tensile strength of the steel was observed to increase with strain rate. Negative strain rate sensitivity of the flow stress in the intermediate temperature range revealed the occurrence of dynamic strain ageing in the steel.
- Voce's constitutive equation was found to describe the tensile flow behaviour of the RAFM steel. Both the yield stress and ultimate tensile strength of the steel could be estimated using Voce's constitutive equation parameters.
- The steel exhibited predominantly stage III strain hardening. Correspondence between tensile flow behaviour and creep deformation behaviour has been established for this microstructural complex steel.

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References

- [1] A. Kohyama, Y. Kohnno, K. Asakura, H. Kayano, J. Nucl. Mater. 212–215 (6) (1994) 84–689.
- [2] N. Baluc, R. Schaublin, P. Spatig, M. Victoria, Nucl. Fission 44 (2004) 56–61.
- [3] G.L. Butterworth, J. Nucl. Mater. 179–181 (1991) 135.
- [4] I. Cook, D. Ward, S. Dudarev, Plasma Phys. Control Fusion 44 (2002) B121–B136.
- [5] R. Jayaram, R.L. Klueh, Metall. Mater. Trans. A 29 (1998) 1551–1558.
- [6] R.L. Klueh, D.J. Alexander, M.A. Sokolov, J. Nucl. Mater. 304 (2002) 139–152.
- [7] P. Fernandez, A.M. Lancha, J. Lapena, M. Hernandez-Mayoral, Fusion Eng. Design 58–59 (2001) 787–792.
- [8] J.H. Hollomon, Trans. AIME 162 (1945) 268–290.
- [9] P. Ludwik, Elemente der Technologischen Mechanik, Springer, Leipzig, 1909. 32p.
- [10] D.C. Ludwigson, Metall. Trans. A 2 (1971) 2825–2828.
- [11] E. Voce, J. Inst. Metal 74 (1948) 537–562.
- [12] B.K. Choudhary, K.B.S. Rao, S.L. Mannan, B.P. Kashyap, Mater. Sci. Technol. 15 (1999) 791–797.
- [13] P. Rodriguez, Bull. Mater. Sci. 6 (4) (1984) 653–663.
- [14] K. Laha, K.S. Chandravathi, K. Bhanu Sankara Rao, S.L. Mannan, Z. Metallkd. 12 (1994) 839–844.
- [15] Huailin Li, A. Nishimura, T. Nagasaka, T. Muroga, Fusion Eng. Des. 81 (2006) 2907–2912.
- [16] C. Girish Shastri, M.D. Mathew, K. Bhanu Sankara Rao, S.L. Mannan, Int. J. Pres. Ves. Piping 81 (2004) 297–301.
- [17] R. Kishore, T.K. Sinha, Metall. Trans. A 27A (1996) 3340–3343.
- [18] Y. Estrin, H. Mecking, Acta Metall. 32 (1984) 57–70.
- [19] H. Mecking, in: Aachen, P. Haasen, V. Gerold, G. Kosterz (Eds.), Proc. 5th Int. Conf. on Strength of Metals and Alloys, Pergamon Press, Oxford, 1979, pp. 231–51.
- [20] Russell.R. Jensen, John.K. Tien, Metal. Trans. A. 16A (1985) 1049–1068.
- [21] M.D. Mathew, J. Vanaja, K. Laha, G. Varaprasad Reddy, K.S. Chandravathi, K. Bhanu Sankara Rao, J. Nucl. Mater. 417 (2011) 77–80.
- [22] J. Vanaja, K. Laha, R. Mythili, K.S. Chandravathi, S. Saroja, M.D. Mathew, Mater. Sci. Eng. A 533 (2012) 17–25.
- [23] G. Gottstein, A.S. Argon, Acta metall. 35 (6) (1987) 1261–1271.
- [24] N.S. Mishra, Sanak Mishra, V. Ramaswamy, Metall. Trans. A 20A (1989) 2819–2829.